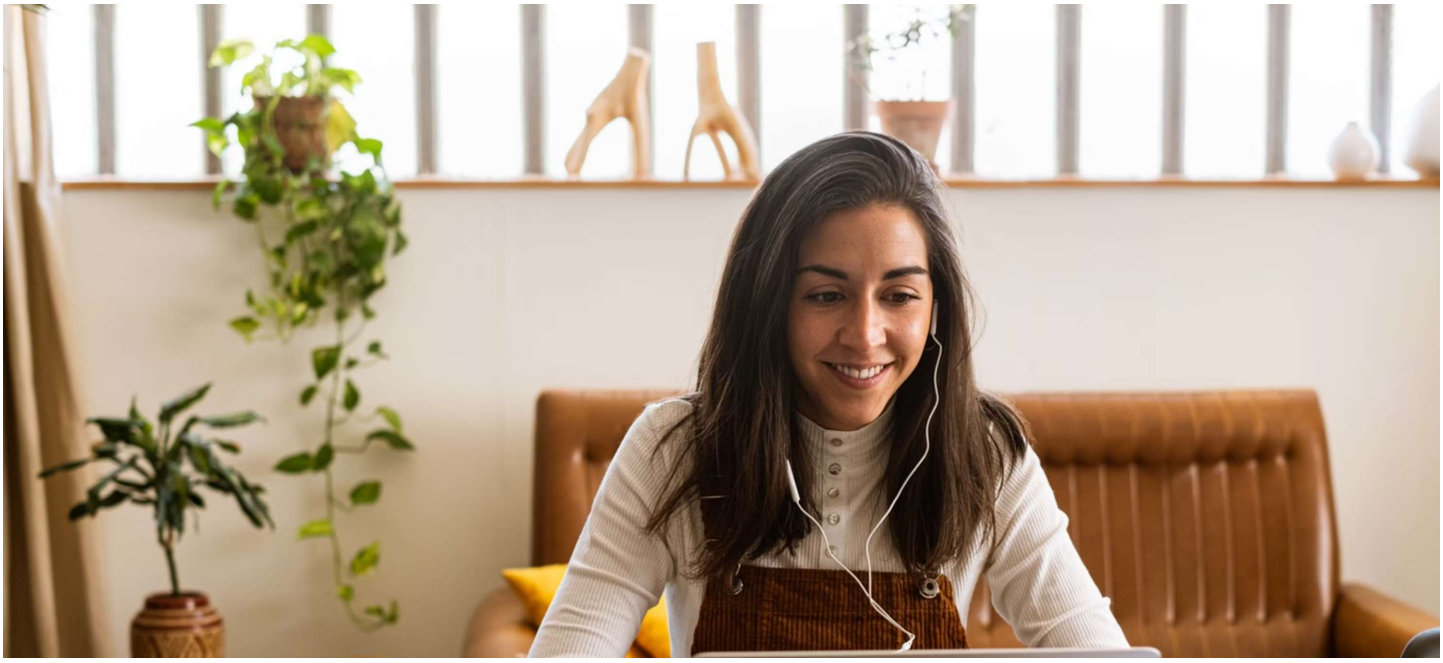
Relational vs. Non-relational Database: The Difference Explained

Автор публикации: Coursera • Обновлено 13 сент. 2023 г.

Поделиться

Relational and non-relational databases are used by businesses to store, retrieve, and analyze data. Learn how these two database types differ from each other.



Whether you're an aspiring data scientist or are simply in job that requires you to use data on a daily basis, you've probably come across the terms **relational** and **non-relational databases**. Both types of databases have clear differences, but you might not be quite sure what they are or how they're used.

Let us guide you through what relational and non-relational databases are, lists of common databases used today, and how to decide which database is best for your needs.

What is a relational database?

A relational database stores data in tables composed of rows and columns.

In a relational database, data is contained within a table, which is then linked to data contained within other tables through the use of unique identifying keys. Specifically, relationships between tables are formed when a primary key, which uniquely identifies a row in one table, connects with a foreign key identifying a row of data in another table.

In effect, relational databases are designed to store **structured data** or well-defined data like names, dates, and quantities that can be easily standardized within a table. SQL, or Structured Query Language, is the most common programming language used to interface with relational databases within relational database management systems (RDMS).

Read more: Relational Database: What It Is and Why It's Important

Common relational databases

Relational databases are suitable for storing, retrieving, and manipulating well-defined, structured data. They are easy to use. Some of the most common relational databases include:

- MySQL
- IBM Db2
- Snowflake
- Amazon Aurora
- PostgreSQL
- Microsoft SQL Server

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A lecture from IBM's Data Engineering Professional Certificate discussing data and relational data bases.

Введение в реляционные базы данных (СУБД)

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Курс 4 из 13 — Инженерия данных IBM Профессиональная сертификация

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What is a non-relational database?

A **non-relational database** is a type of database that doesn't store data in tables but instead in whatever format is best for the type of data being stored. In effect, non-relational databases are designed to contain **unstructured data**, or loosely defined data like email messages, videos, images, and business documents that aren't easily standardized. They can also be used to store a mix of structured and unstructured data.

Non-relational databases are said to be NoSQL, meaning that they don't use Structured Query Language, even though many NoSQL databases do support SQL queries.

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Types of non-relational databases

Unlike relational databases, which only support a tabular design, there are many different types of non-relational databases. Some of the most common approaches to organizing non-relational databases include:

- **Key-value stores:** In a key-value store, data is assigned a unique identifier, which allows it to be retrieved and sorted. The system consists of "keys," or unique identifiers like a string of numbers, and the "values" with which they're associated, which can be either data itself or simply its location within the database.

- **Column-family data stores:** In a column-family data store, data is organized in a "keyspace" containing multiple families of different columns. The column families, in turn, include rows that each also contain columns containing data. The data can be accessed and identified using keys.

- **Graph databases:** Graph databases store data in nodes and structures them based on their relationships to one another, allowing for a much more flexible schema than can be found in tabular designs.

- **Document databases:** Document databases store data within documents, which typically contain one object and all its associated metadata.

Common non-relational databases

Some of the most common non-relational databases include:

- MongoDB
- IBM Cloundant
- Amazon DynamoDB
- Apache Cassandra

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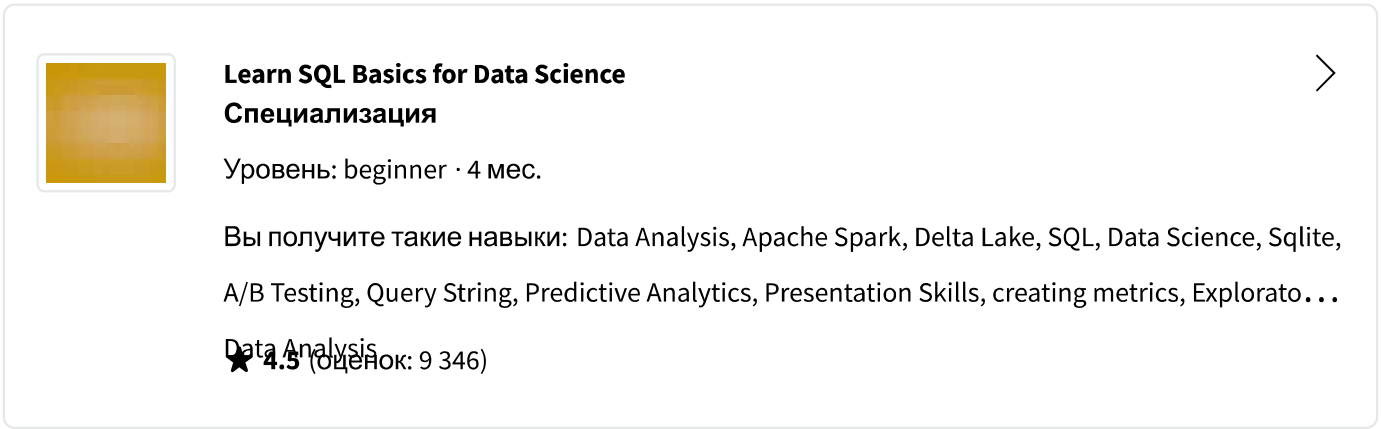
SQL vs NoSQL: What's the difference?

Structured query language (SQL) is the language used in relational databases and is used for *structured data*. **NoSQL** stands for "**not only structured query language**" and is used for non-relational databases, which store both structured *and* unstructured data.

In short, SQL databases are relational databases, and NoSQL databases are non-relational databases.

Read more: NoSQL vs. SQL Databases: Understand the Differences and When to Use

If you want to learn more, the Learn SQL Basics for Data Science Specialization from UC Davis teaches beginners with no previous coding experience the basics of SQL, data wrangling, AB testing, and more.



When to use a relational vs. a non-relational database

One question you might ask is: "When should I use a relational database, and when should I use a non-relational database?"

Here are some considerations to help guide your decision on the appropriate database to meet your needs:

1. Are you working with structured or unstructured data?

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The most important factor to consider when deciding whether you should use a relational or non-relational database is the type of data you are storing. If you're working with structured data that is easily stored within tables, then you will likely be using a relational database. If you're working with either unstructured data or a mix of both structured and unstructured data, then you will need to use a non-relational database.

Note: While a non-relational database can store structured data, a relational database cannot store unstructured data.

2. How much data are you storing?

Another factor impacting the kind of database you decide to use is the volume of data you will store in your database. Typically, non-relational databases are better suited to large volumes of data than relational databases, which work best with small to medium-sized amounts of data.

If you're working with big data or ever-growing amounts of real-time data, then consider using a non-relational database.

3. Who is managing your data and for what purpose?

The last question to consider is who will be using your database regularly and for what purpose.

Generally speaking, non-professionals find that relational databases are much easier to use than non-relational databases, which can require additional training. However, there are some scenarios when a non-relational database makes more sense for your needs, such as if you are continuously collecting real-time business data that must also be quickly analyzed to form business insights.

The right database is the one that best fits your business and its goals, so make sure to consider these important questions before picking a database.

Learn data science with IBM

Databases help businesses store, retrieve, and analyze key data points in order to gain valuable insights. Prepare for a career working with data today by taking a cost-effective online course through Coursera.

IBM's Introduction to Data Science Specialization teaches concepts like big data, statistical analysis, and relational databases. You'll gain familiarity with various open-source tools and data science programs used by data scientists, like Jupyter Notebooks, RStudio, GitHub, and SQL.

In IBM's Data Engineering Professional Certificate, meanwhile, you'll develop working knowledge of NoSQL and Big Data using MongoDB, Cassandra, Cloudant, Hadoop, Apache Spark, Spark SQL, Spark ML, and Spark Streaming. You'll also learn to create, design, and manage relational databases.



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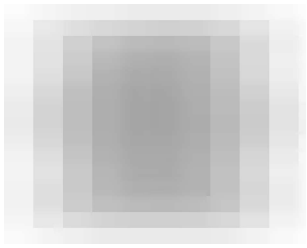
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